

ENEL445 — Worked Examples: Quadratic Optimisation & Optimality Conditions

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Q1. Design Variable Transformation via Sigmoid

[5 marks]

Problem. The i -th design variable x_i must satisfy $|x_i| \leq a$. Eliminate this constraint using the sigmoid $\sigma(t) = \frac{1}{1 + e^{-t}}$.

Dr. Yang: "This can be equivalently transformed to: x must lie in $[-a, a]$. Width of that interval is $2a$." (0:00:56–0:01:07)

Step 1 — restate the constraint.

$|x_i| \leq a$ is equivalent to $x_i \in [-a, a]$, an interval of width $2a$.

$$-a < x_i < +a$$

Dr. Yang: "This sigmoid function takes values from zero to one ... the range has width $2a$, so we can scale it." (0:01:07–0:01:40)

Step 2 — range of the sigmoid.

$$\sigma(t) = \frac{1}{1 + e^{-t}}, \quad t \in \mathbb{R} \Rightarrow \sigma(t) \in (0, 1).$$

Reason: If $t \rightarrow +\infty$, $e^{-t} \rightarrow 0$ so $\sigma \rightarrow 1$. If $t \rightarrow -\infty$, $e^{-t} \rightarrow \infty$ so $\sigma \rightarrow 0$. The image is strictly $(0, 1)$ for all finite t .

Dr. Yang: "This part $(2a/(1 + e^{-t}))$ gives you zero to a ... if you add $-a$, the whole thing becomes $-a$. That is the sanity check. Two-line answer." (0:02:02–0:02:20)

Step 3 — scale and shift to $(-a, a)$.

Multiply by $2a$ to obtain range $(0, 2a)$, then shift by $-a$:

$$\begin{aligned} 2a\sigma(t_i) &\in (0, 2a) \\ x_i = -a + 2a\sigma(t_i) &\in (-a, +a). \end{aligned}$$

Reason: Sanity check: $t_i \rightarrow -\infty \Rightarrow x_i = -a + 0 = -a$. $t_i \rightarrow +\infty \Rightarrow x_i = -a + 2a = +a$. For all finite t_i , $x_i \in (-a, +a) \subseteq [-a, a]$. ✓

Step 4 — explicit bound confirmation.

For all finite $t_i \in \mathbb{R}$, the transformed variable satisfies:

$$x_i \in (-a, +a).$$

The open interval $(-a, +a)$ is **always** a subset of the closed constraint set:

$$(-a, +a) \subseteq [-a, a]. \checkmark$$

Reason: The sigmoid never reaches its limits for finite t_i , so the image is strictly the open interval $(-a, +a)$, giving margin on both sides. The constraint is therefore satisfied automatically and can be dropped entirely from the optimisation.

Why use $\sigma(t)$?

The sigmoid is one member of a family of functions that share three key properties required for this variable-transformation trick:

1. **Bounded range** — maps $\mathbb{R}$ into a finite interval, so the constraint on x_i is encoded implicitly.
2. **Smooth and differentiable everywhere** — so the chain rule $\partial f / \partial t_i = (\partial f / \partial x_i)(\partial x_i / \partial t_i)$ is well defined, and gradient-based optimisers can be used on t_i without modification.
3. **Strictly monotone** — the mapping $t_i \mapsto x_i$ is one-to-one, so no information is lost and every $x_i \in (-a, a)$ has a unique pre-image.

Reason: Any smooth, strictly monotone function with bounded range works. Alternatives include $\tanh(t)$ (range $(-1, 1)$, scaled by a) and $\frac{2}{\pi} \arctan(t)$ (range $(-1, 1)$). The sigmoid is convenient because its range is exactly $(0, 1)$, which maps cleanly to $(-a, a)$ with one multiply-and-shift. Without this trick, a constrained method (e.g. KKT conditions) would be needed; the transformation converts the problem into an unconstrained one solvable by any standard solver.

Substitute $x_i = -a + \frac{2a}{1 + e^{-t_i}}$ and optimise over unconstrained $t_i \in \mathbb{R}$. The constraint $|x_i| \leq a$ is then automatically satisfied and can be dropped.

Equivalent forms: $x_i = a \left(\frac{2}{1 + e^{-t_i}} - 1 \right) = a \tanh\left(\frac{t_i}{2}\right)$.

Q2. First-Order Necessary Optimality Condition

[5 marks]

Problem. Show that if x^* is a local minimum of $f(x)$, then $\nabla f(x^*) = 0$. *Hint: use the first-order Taylor expansion of $f(x)$.*

: $d \in \mathbb{R}^n$ is introduced as an arbitrary perturbation vector (a direction and magnitude we are free to choose later), and $\delta > 0$ is the radius of the neighbourhood in which x^* is a minimum.

Step 1 — local-minimum definition. There exists $\delta > 0$ such that

$$f(x^* + d) \geq f(x^*) \quad \text{for all } \|d\| < \delta.$$

: This neighbourhood is written $\mathcal{N}(x^*, \delta) = \{x^* + d \mid \|d\| < \delta\}$ — the set of all points within distance δ of x^* . The δ -ball condition says: every point in $\mathcal{N}(x^*, \delta)$ must have f -value at least as large as $f(x^*)$. This is the inequality we will contradict if $\nabla f(x^*) \neq 0$.

Step 2 — first-order Taylor expansion about x^* .

$$\begin{aligned} f(x) &= f(x^*) + \nabla f(x^*)^\top (x - x^*) + O(\|x - x^*\|^2) \\ f(x^* + d) &= f(x^*) + \nabla f(x^*)^\top d + O(\|d\|^2). \end{aligned}$$

Reason: The chain rule gives the linear term $\nabla f(x^*)^\top d$; higher-order terms are $O(\|d\|^2)$.

: Substitute the Taylor expansion into the local-minimum inequality, then cancel $f(x^*)$ from both sides to isolate the gradient term.

Step 3 — substitute into the local-minimum inequality.

Substituting the Taylor expansion into $f(x^* + d) \geq f(x^*)$:

$$\begin{aligned} f(x^*) + \nabla f(x^*)^\top d + O(\|d\|^2) &\geq f(x^*) \\ \nabla f(x^*)^\top d + O(\|d\|^2) &\geq 0 \quad \text{for all } \|d\| < \delta. \end{aligned}$$

Dr. Yang: "If the gradient is not equal to zero, we can always find a direction d that can further decrease the cost function. Therefore the gradient at this particular point must equal zero." (0:03:24–0:03:47)

: The direction $d = -\varepsilon \nabla f(x^*)$ is the steepest-descent direction scaled by ε . If $\nabla f(x^*) \neq 0$, this direction would decrease f for small enough ε , contradicting Step 3.

Step 4 — choose a test direction (contradiction). Set $d = -\varepsilon \nabla f(x^*)$ with $\varepsilon > 0$ small:

$$-\varepsilon \|\nabla f(x^*)\|^2 + O(\varepsilon^2) \geq 0.$$

: Since $\varepsilon > 0$, dividing preserves the direction of the inequality. Letting $\varepsilon \rightarrow 0^+$ kills the $O(\varepsilon)$ remainder, isolating the term involving $\|\nabla f(x^*)\|^2$.

Step 5 — divide by ε and let $\varepsilon \rightarrow 0^+$.

Reason: Dividing by $\varepsilon > 0$ preserves the inequality; the $O(\varepsilon)$ term vanishes.

$$\begin{aligned} -\|\nabla f(x^*)\|^2 &\geq 0 \\ \|\nabla f(x^*)\|^2 &\leq 0 \end{aligned}$$

But a squared norm is **always** non-negative:

$$\|\nabla f(x^*)\|^2 \geq 0$$

$$0 \leq \|\nabla f(x^*)\|^2 \leq 0 \implies \|\nabla f(x^*)\|^2 = 0 \implies \nabla f(x^*) = 0. \square$$

Q3. Van der Corput Sequence, base $b = 3$

[10 marks]

Definition. If $i = a_0 + a_1b + a_2b^2 + \dots + a_rb^r$ in base b , then

$$\phi_i(b) = \frac{a_0}{b} + \frac{a_1}{b^2} + \dots + \frac{a_r}{b^{r+1}}.$$

: The Van der Corput sequence is a *low-discrepancy* sequence: it fills $[0, 1]$ more uniformly than random sampling. After b^k terms, the points are spaced exactly $1/b^k$ apart. It is used in quasi-Monte Carlo methods and space-filling designs for optimisation.

Dr. Yang: "For $i = 1$, the representation in base 3 is $(1)_3$, so the sequence value is $1/3$. For $i = 3$, that is $(10)_3$, giving $1/9$. And so on — simple." (0:03:53–0:05:18)

Algorithm (three steps for each i):

1. Write i in base b , reading off digits $a_0, a_1, \dots, a_r$ least-significant first.
2. Each digit a_k contributes a_k/b^{k+1} to the sum.
3. Sum all contributions: $\phi_i(b) = \sum_{k=0}^r \frac{a_k}{b^{k+1}}.$

: The key bridging fact: the formula places the digits of i **after** the radix point in base b , in reverse order — the least-significant digit of i becomes the most-significant fractional digit. For example, $i = 3 = (10)_3$ has $a_0 = 0, a_1 = 1$, giving $\phi_3(3) = 0.01_3 = 0/3 + 1/9 = 1/9$. This reversal is what spreads successive points across $[0, 1]$.

i	Base-3 repr.	Digits $a_0, a_1, \dots$	Computation	$\phi_i(3)$
1	$(1)_3$	$a_0 = 1$	$1/3$	$1/3$
2	$(2)_3$	$a_0 = 2$	$2/3$	$2/3$
3	$(10)_3$	$a_0 = 0, a_1 = 1$	$0/3 + 1/9$	$1/9$
4	$(11)_3$	$a_0 = 1, a_1 = 1$	$1/3 + 1/9$	$4/9$
5	$(12)_3$	$a_0 = 2, a_1 = 1$	$2/3 + 1/9$	$7/9$

Reason: $i = 1$: $(1)_3, a_0 = 1 \Rightarrow 1/3$. $i = 2$: $(2)_3, a_0 = 2 \Rightarrow 2/3$.

$i = 3$: $3 = (1 \cdot 3 + 0) = (10)_3, a_0 = 0, a_1 = 1 \Rightarrow 0/3 + 1/9 = 1/9$.

$i = 4$: $4 = (1 \cdot 3 + 1) = (11)_3, a_0 = 1, a_1 = 1 \Rightarrow 1/3 + 1/9 = 4/9$.

$i = 5$: $5 = (1 \cdot 3 + 2) = (12)_3, a_0 = 2, a_1 = 1 \Rightarrow 2/3 + 1/9 = 7/9$.

Van der Corput sequence in base $b = 3$ for $i = 1, \dots, 5$:

$$\phi_1(3) = \frac{1}{3}$$

$$\phi_2(3) = \frac{2}{3}$$

$$\phi_3(3) = \frac{1}{9}$$

$$\phi_4(3) = \frac{4}{9}$$

$$\phi_5(3) = \frac{7}{9}.$$

Q4. Gradient of Quadratic Objective

[3 marks]

Problem. Let $f(x) = \frac{1}{2}x^\top Ax + b^\top x$, A symmetric positive definite. Find $\nabla f(c)$.

Dr. Yang: “Most of you did this well. The gradient is $Ac + b$. Some of you wrote $b^\top$ — the gradient is a *vector*, not a scalar.” (0:05:21–0:05:35)

Linear term: $\partial(b^\top x)/\partial x = b$.

Reason: Standard result: $\partial(a^\top x)/\partial x = a$.

Quadratic term (using $A = A^\top$):

$$\frac{\partial}{\partial x} \left(\frac{1}{2} x^\top A x \right) = \frac{1}{2} (A + A^\top) x = Ax.$$

Reason: Product rule gives $(A + A^\top)x/2$; symmetry collapses this to Ax .

$$\nabla f(x) = Ax + b, \quad \text{so} \quad \nabla f(c) = Ac + b.$$

Q5. Exact Line Search Formulation

[5 marks]

Problem. Formulate the exact line search at c along $-\nabla f(c)$ as an unconstrained optimisation in μ .

Dr. Yang: “The answer is one line. Step size must be ≥ 0 , and we move from c in direction $-\nabla f(c)$.” (0:05:36–0:06:11)

$$\min_{\mu \geq 0} \phi(\mu) = f(c - \mu \nabla f(c)).$$

Reason: We are at the current point c and search along the steepest-descent direction $-\nabla f(c)$. The scalar $\mu \geq 0$ controls how far we step. For a quadratic with $A \succ 0$, $\phi(\mu)$ is convex with a unique minimiser $\mu^* > 0$, so the constraint is automatically satisfied.

Q6. Optimal Step Size μ^*

[12 marks]

Problem. Solve the above for μ^* and write the updated point $d = c - \mu^* \nabla f(c)$.

Dr. Yang: “We differentiate with respect to the step size and set the answer to zero. That is the chain rule — we have covered this in lectures.” (0:06:41–0:06:56)

Let $g = \nabla f(c) = Ac + b$ (from Q4).

Step 1 — differentiate $\phi(\mu)$ via chain rule:

$$\begin{aligned}\frac{d\phi}{d\mu} &= (\nabla f(c - \mu g))^\top \cdot \frac{d(c - \mu g)}{d\mu} \\ &= (\nabla f(c - \mu g))^\top \cdot (-g) \\ &= -\nabla f(c - \mu g)^\top g = 0.\end{aligned}$$

Reason: Chain rule: derivative of the outer function (∇f evaluated at $c - \mu g$) times derivative of the inner function ($-g$).

: The inner function $c - \mu g$ is linear in μ , so $d(c - \mu g)/d\mu = -g$. The outer function is f , so the chain rule gives ∇f evaluated at the current argument $c - \mu g$.

Dr. Yang: “You already know this part from the gradient question.” (0:07:29–0:07:33)

Step 2 — evaluate $\nabla f(c - \mu g)$ using the result from Q4:

$$\nabla f(c - \mu g) = A(c - \mu g) + b = (Ac + b) - \mu Ag = g - \mu Ag.$$

Step 3 — set to zero:

$$\begin{aligned}-(g - \mu Ag)^\top g &= 0 \\ -g^\top g + \mu g^\top Ag &= 0 \\ \mu^* &= \frac{g^\top g}{g^\top Ag}.\end{aligned}$$

Reason: Expand: $-g^\top g + \mu g^\top Ag = 0$. Since $A \succ 0$, $g^\top Ag > 0$, so we can divide through.

Dr. Yang: “The final answer: $\mu^* = \nabla f(c)^\top \nabla f(c) / \nabla f(c)^\top A \nabla f(c)$.” (0:07:33–0:07:55). The asterisk denotes the *optimal* (minimising) value.

$$\mu^* = \frac{\|\nabla f(c)\|^2}{\nabla f(c)^\top A \nabla f(c)}, \quad d = c - \frac{\|\nabla f(c)\|^2}{\nabla f(c)^\top A \nabla f(c)} \nabla f(c).$$

This is the exact steepest-descent step size for a quadratic objective.